

Problem–Solution Fit Canvas

Date: 29 june 2026

Team ID: (xxxxxx)

Project Name: Human Development Index (HDI) Prediction Using Machine Learning

1. Customer Segment(s) [CS]

- Government Policy Makers
 - Researchers
 - International Development Organizations
 - Students and Educators
 - Economists and Data Analysts
 - NGOs working on social and economic development
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2. Problems / Pains + Its Frequency [PR]

- Manual HDI analysis is time-consuming.
- Comparing multiple countries is difficult.
- Large socio-economic datasets are complex to analyze.
- Identifying development gaps requires expert knowledge.
- Traditional methods are slow and prone to human error.

Frequency: High (used regularly for policy planning, research, and education).

3. Triggers to Act [TR]

- Need to predict the HDI category quickly.

□

- Compare development levels of countries.
 - Support policy and development planning.
 - Analyze the impact of health, education, and income indicators.
 - Generate data-driven insights for decision-making.
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4. Emotions Before / After [EM]

Before Using the System

Confused by complex datasets.

- Concerned about manual calculations.
- Unsure of prediction accuracy.

After Using the System

Confident in prediction results.

- Satisfied with quick analysis.
 - Better informed for planning and decision-making.
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5. Available Solutions – Pros & Cons [AS] Existing Solutions

- Manual spreadsheet analysis
- Statistical reports
- UNDP HDI reports
- Traditional data analysis tools

Pros

- Reliable historical information.
- Official development statistics.
- Widely accepted data sources.

□

Cons

- Time-consuming.
 - No instant prediction capability.
 - Requires expert knowledge for interpretation.
 - Limited interactivity.
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6. Customer Limitations [CL]

- Limited technical knowledge.
 - Limited access to advanced analytical tools.
 - Time constraints.
 - Budget limitations.
 - Difficulty interpreting complex socio-economic data.
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7. Behavior + Its Intensity [BE]

- Frequently compares country development indicators.
- Searches for HDI reports and research papers. □ Uses statistical data for planning and analysis.
- Requires quick and accurate prediction results.

Intensity: High

8. Channels of Behavior [CH] Online

- Flask Web Application
- GitHub Repository
- Government and UNDP websites
- Educational portals

□

- Research databases

Offline

Universities

Government offices

Research institutions

□ Conferences and workshops

9. Problem Root / Cause [RC]

- HDI depends on multiple socio-economic indicators that are difficult to evaluate manually.
 - Manual calculations are slow and may lead to errors.
 - Lack of an intelligent prediction system for quick analysis.
 - Existing tools mainly provide historical data instead of predictive insights.
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Team Member	Role	Responsibility
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G Naveen	Team	Project planning, GitHub management, and model
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10. Your Solution [SL]

The proposed solution is a **Machine Learning-based Human Development Index Prediction System**.

The system:

- Accepts Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI Per Capita as inputs.
- Uses a trained Machine Learning model to predict the HDI category.
- Classifies countries into **Very High, High, Medium, or Low** HDI.
- Provides instant and accurate predictions through a Flask-based web application.
- Helps governments, researchers, students, and policymakers make informed decisions and identify development priorities.

Team Members and Responsibilities :-

Team Member	Role	Responsibility
	Lead	integration
Chandana Busireddy	Member	Dataset collection, preprocessing, and documentation
Gandham Joshna	Member	Data visualization and exploratory data analysis
Nandineni Gowthami	Member	Flask web application development and testing

Shaik Talha	Member	Machine learning model training, report preparation, GitHub updates, and presentation
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